

SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY

(Department of Computer Science and Engineering)

Format for Minor Project Report

SPECIFICATIONS FOR MINOR PROJECT REPORT

1. Report shall be computer typed (English- British, Font -Times New Roman, Size- 12 point) and printed on A4 size paper.
2. The Report shall be in prescribed format, one sided printed , in sessional file.
3. The report shall be typed on one side only with double space with a margin of 3.5 cm on the left, 2.5 cm on the top, and 1.25 cm on the right and at bottom and all text ,must be justified.
4. In the report, the title page [Refer sample sheet] should be given first then the Certificate by the candidate and the supervisor(s) in sequence, followed by Approval Certificate, followed by Declaration and then an acknowledgment. This should be followed by the abstract of the report (not exceeding 1500 words), list of figures/list of tables, notations/nomenclature, and then contents with page no.s
5. References and Bibliography should be included in report.
7. The diagrams should be printed on a light/white background, Tabular matter should be clearly arranged. Decimal point may be indicated by full stop(.)The caption for Figure must be given at the BOTTOM of the Fig. and Caption for the Table must be given at the TOP of the Table.
8. The graphs should be combined for the same parameters for proper comparison.

Single graph should be avoided as far as possible.
9. Conclusions must not exceed more than two pages.
10. The report must consist of the following chapters

Chapter 1- Introduction

Chapter 2- Requirement Analysis and System Specification

Chapter 3- System Design

Chapter 4- Implementation, Testing and Maintenance

Chapter 5-Results and Discussions Chapter

6-Conclusion and Future Scope References

Appendix (if any)

Annexures-I,II,III

11. There should be 2 copies of the Final Report for submission in the department and you can create separate copies for each team member.

12. Report should contain min 70 pages

QUIZ CRAFT AI – AI BASED QUIZ GENRATOR SHUFFLE BASED

MINOR PROJECT REPORT

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY
(Computer Science and Engineering)



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SAGAR NSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL(M.P.)

Department of Computer Science and Engineering



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, Bhopal

Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the Minor Project “**Quiz Craft AI**” being Submitted by **Pranshu Mishra [0133CS231235], PriyaRanjan Singh [0133CS231243]**, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** to **RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL (M.P.)** during the academic year **2025-26** is a record of bona fide piece of work, carried out by her under my supervision and guidance in the **Department of Computer Science and Engineering, Sagar Institute of Research & Technology, Bhopal.**

Guide Name
CSE Dept.
Supervisor

Dr. Mohit Singh Tomar
HOD CSE

Prof. Arun Kumar Jhapate
HOD CSE



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, Bhopal
Department of Computer Science & Engineering

APPROVAL CERTIFICATE

The Minor Project entitled “Quiz Craft AI” submitted by “Pranshu Mishra[0133CS231235], PriyaRanjan Singh [0133CS231243]” to Rajiv Gandhi Proudhyogiki Vishwavidyalaya in partial fulfillment of the requirements for the Minor Project evaluation of the pre-final year Bachelor of Technology in Computer Science and Engineering during the academic session 2025–26 has been examined by us and is hereby approved.

(Internal Examiner)

(External Examiner)

Date:

Date:



SAGAR NSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL

Department of Computer Science & Engineering

DECLARATION

I hereby declare that the work, which is being presented in the Minor project, entitled " *Quiz Craft AI*" Submitted in the **Department of Computer Science & Engineering, Sagar Institute of Research & Technology, Bhopal** is an authentic record of my own work carried out during the period from May 2025 - June 2026, under the guidance of "***Prof.Praveen Kaithal***" **Department of Computer Science & Engineering, Sagar Institute of Research & Technology, Bhopal.**

Date: 03-06-2026

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Place: Bhopal



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, Bhopal

Department of Computer Science & Engineering

ACKNOWLEDGEMENT

This Minor Project marks one of the most memorable milestones of my B.Tech. program. I take this opportunity to express my sincere gratitude to all those who have directly or indirectly contributed to the successful completion of this work.

I express my heartfelt thanks to **Prof. Praveen Kaithal** for providing valuable guidance, constant encouragement, and continuous support throughout the project work. Their insightful suggestions, motivation, and supervision helped me complete this project successfully.

I am grateful to the faculty members of the Department of Computer Science and Engineering for their support and encouragement during the course of this project.

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I am thankful to all the staff members of the CSE Department and my friends for their timely help, cooperation, and valuable suggestions during the project work.

Finally, I express my deepest gratitude to my family for their unconditional love, encouragement, and moral support, without which the successful completion of this Minor Project would not have been possible.

Pranshu Mishra

PriyaRanjan Singh

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Abstract

The rapid growth of digital education and e-learning platforms has increased the demand for intelligent assessment systems capable of evaluating students efficiently and accurately. Traditional quiz creation methods often require significant manual effort from educators, making the process time-consuming, repetitive, and difficult to manage. Furthermore, many existing quiz systems rely on fixed question patterns, reducing assessment fairness and limiting student engagement.

This project presents **Quiz Craft AI**, an AI-powered MCQ Quiz Generator that automates the creation, management, and evaluation of quizzes using modern web technologies and Artificial Intelligence. The proposed system integrates **HTML, CSS, JavaScript, Python FastAPI, and MongoDB** to provide an intelligent platform for generating multiple-choice questions from educational content and study materials. Unlike traditional quiz systems that depend on manually created question banks, Quiz Craft AI utilizes AI-driven content analysis to automatically generate relevant and meaningful assessment questions.

The system processes uploaded study materials, identifies important concepts, and generates multiple-choice questions along with answer options and correct answers. It also supports automatic question randomization, quiz management, instant score calculation, and performance evaluation. Generated quizzes and assessment records are securely stored in a MongoDB database, allowing users to maintain quiz history and monitor learning progress through interactive dashboards and performance analytics.

The proposed solution significantly reduces the effort required for manual quiz preparation while improving the efficiency and reliability of online assessments. By combining AI-powered content analysis, automated question generation, quiz evaluation, and performance tracking, the system transforms traditional assessment methods into a more intelligent and interactive learning experience. Experimental evaluation demonstrates that the platform efficiently generates quizzes, improves assessment management, and enhances student engagement.

Thus, Quiz Craft AI serves as an intelligent educational assistant that bridges the gap between traditional assessment methods and modern AI-driven learning technologies, enabling educators

to create assessments more efficiently while helping students evaluate and improve their
knowledge effectively.

Keywords: Artificial Intelligence, Quiz Generation, MCQ Generator, FastAPI, MongoDB, Educational Technology, Online Assessment, E-Learning, Question Randomization, Digital Learning.

Chapter 1 Introduction

1.1 Background

1.2 Project Category(Internet based, Application or System Development, Research based ,Industry Automation, Network or System Administration)

1.3 Objectives

1.4 Problem Formulation

1.5 Identification/Reorganization of Need

1.6 Existing System

1.7 Proposed System

1.8 Unique Features of the System

Chapter 2. Requirement Analysis and System Specification

2.1 Feasibility study (Technical, Economical, Operational)

2.2 Software Requirement Specification Document which must include the following:

(Data Requirement, Functional Requirement, Performance Requirement, Dependability Requirement, Maintainability requirement, Security requirement, Look and feel requirement)

2.3 Validation

2.4 Expected hurdles

2.5 SDLC model to be used

Chapter 3. System Design

3.1 Design Approach (Function oriented or Object oriented)

3.2 Detail Design

3.3 System Design using various Structured analysis and design tools such as: DFD's, Data Dictionary, Structured charts, Flowcharts or UML

3.4 User Interface Design

3.5 Database Design3.5.1 ER Diagrams

3.5.2 Normalization

3.5.3 Database Manipulation

3.5.4 Database Connection Controls and Strings

3.6 Methodology

Chapter 4. Implementation, Testing, and Maintenance

4.1 Introduction to Languages, IDE's, Tools and Technologies used for Implementation

4.2 Coding standards of Language used

4.3 Project Scheduling using various tools such as PERT, GANTT charts, Open PROJ etc.

4.4 Testing Techniques and Test Plans

Chapter 5. Results and Discussions

5.1 User Interface Representation (of Respective Project)

5.1.1 Brief Description of Various Modules of the system

5.2 Snapshots of system with brief detail of each

5.3 Back Ends Representation (Database to be used)

5.3.1 Snapshots of Database Tables with brief description

Chapter 6. Conclusion and Future Scope

Note: The report of respective project should be as per prescribed format and in the same order though if some of the points are not applicable in regard with the project concerned, they might be omitted.

CHAPTER 1

INTRODUCTION

1.1 Background

Functional requirements define what the system must do.

User Authentication Module

The system shall:

- Allow user registration.
- Allow secure login and logout.
- Maintain authenticated user sessions.

Content Upload Module

The system shall:

- Accept PDF and educational documents.
- Validate file type and size.
- Upload content securely to the server.

AI Quiz Generation Module

The system shall:

- Process uploaded study materials.
- Extract important educational information.
- Generate multiple-choice questions automatically.
- Create answer options and identify correct answers.

Quiz Management Module

The system shall:

- Create quizzes dynamically.
- Shuffle questions and answer options.
- Manage quiz attempts.
- Store quiz-related information.

Evaluation Module

The system shall:

- Evaluate user responses automatically.
- Calculate quiz scores.
- Display assessment results instantly.
- Generate performance summaries.

Dashboard Module

The system shall:

- Display quiz history.
- Show performance statistics.
- Visualize assessment results.
- Present learning progress information.

History Module

The system shall:

- Store previous quiz attempts.
- Allow retrieval of past assessments.

- Maintain long-term performance records.

1.2 Project Category

The proposed project falls under multiple categories:

Internet-Based Application

Quiz Craft AI is primarily an internet-based application because it utilizes web technologies and online services to generate, manage, and conduct quizzes. Users can access the platform through a web browser, upload study materials, generate quizzes, and participate in assessments from any location with internet connectivity.

Application Development Project

The project involves the design and development of a complete web application consisting of frontend, backend, database, and AI integration modules. The system provides an interactive environment for both students and educators, enabling efficient quiz generation and assessment management.

Research-Based Project

The project incorporates modern Artificial Intelligence and Natural Language Processing (NLP) techniques for automatic MCQ generation from educational content. It explores the practical application of AI in the education sector and demonstrates how intelligent systems can improve learning and assessment processes.

System Development Project

Quiz Craft AI functions as a complete educational assessment system that generates, stores, manages, and evaluates quizzes. The system integrates multiple software components to provide an end-to-end solution for online learning and knowledge evaluation.

Educational Technology Project

The project also belongs to the Educational Technology (EdTech) domain as it aims to enhance digital learning experiences through intelligent automation, online assessments, and AI-powered educational tools.

Therefore, the project can be classified as:

Internet-Based Web Application with AI-Driven Quiz Generation, System Development, and Educational Technology Components.

1.3 Objectives

The primary objective of **Quiz Craft AI** is to automate the process of quiz creation and improve the effectiveness of online assessments using Artificial Intelligence. The system aims to reduce the manual effort involved in quiz preparation while providing students with a more engaging and fair learning experience.

Main Objectives

To develop a web-based platform for AI-powered MCQ quiz generation.

To automatically generate multiple-choice questions from educational content and study materials.

To reduce the time and effort required for manual quiz creation.

To provide randomized question and option shuffling for fair assessments.

To enable students to participate in online quizzes and evaluate their knowledge effectively.

To store generated questions, quiz records, and assessment results in a database.

To provide instant quiz evaluation and score calculation.

To improve learning outcomes through self-assessment and practice-based learning.

To create a user-friendly interface for both students and educators.

To enhance the reliability and transparency of online examinations.

To demonstrate the practical application of Artificial Intelligence in educational technology.

To develop a scalable and flexible architecture that can support future enhancements such as adaptive learning, performance analytics, and personalized quiz recommendations.

To support educational institutions, coaching centers, and online learning platforms with an efficient digital assessment solution.

To improve the overall quality, accessibility, and efficiency of technology-driven learning and evaluation systems.

1.4 Problem Formulation

The increasing adoption of online learning and digital education platforms has created a growing need for efficient and intelligent assessment systems. However, traditional quiz creation methods still rely heavily on manual efforts, making the process time-consuming, repetitive, and difficult to manage. Educators often spend significant time preparing question banks, organizing quizzes, and evaluating student performance.

Many existing quiz systems follow fixed question patterns and provide the same set of questions to all users. This makes assessments predictable and increases the possibility of answer sharing among students, especially in online examinations. As a result, the fairness and reliability of the evaluation process may be compromised.

The primary problems identified are:

- Time-consuming manual quiz creation and management.
- Fixed question patterns that make quizzes predictable.
- Increased chances of cheating and answer sharing.
- Lack of intelligent automation in question generation.
- Difficulty in creating large numbers of quality MCQs.
- Limited support for personalized and dynamic assessments.
- Increased workload for educators and instructors.
- Lack of efficient tools for self-assessment and practice-based learning.

- Repetitive quizzes that reduce student engagement.
- Inefficient management of educational content and question banks.

These challenges create a need for an intelligent system capable of automatically generating multiple-choice questions, managing quizzes efficiently, and providing a fair and engaging assessment environment. Quiz Craft AI is designed to address these issues by utilizing Artificial Intelligence to automate quiz generation, improve assessment quality, and enhance the overall learning experience for both students and educators

1.5 Identification / Recognition of Need

The need for **Quiz Craft AI** emerged from the increasing adoption of online learning platforms and the challenges associated with traditional quiz creation methods. Educators often spend significant time preparing assessments, while students require more interactive and effective ways to evaluate their knowledge.

The following needs were identified:

Need for Automated Quiz Generation

Teachers and instructors require a system that can automatically generate multiple-choice questions from educational content, reducing the time and effort involved in manual quiz preparation.

Need for Fair and Dynamic Assessments

Traditional quiz systems often present the same set of questions to all users, making assessments predictable. There is a need for randomized and dynamic quizzes that ensure fairness and reduce the chances of answer sharing.

Need for Efficient Learning Evaluation

Students need a quick and reliable method to assess their understanding of various subjects through practice quizzes and self-evaluation tools.

Need for Reduced Instructor Workload

Educational institutions require solutions that simplify quiz management and allow instructors to focus more on teaching activities rather than administrative tasks.

Need for User-Friendly Digital Assessment Platforms

Both students and educators require an intuitive and accessible platform that can be used without extensive technical knowledge.

Need for AI-Assisted Educational Solutions

Modern Artificial Intelligence technologies provide opportunities to automate question generation, improve assessment quality, and enhance learning experiences through intelligent educational tools.

Need for Scalable Online Learning Systems

As digital education continues to grow, there is a need for scalable systems that can support multiple users, large question banks, and future feature enhancements.

The identified needs directly motivated the development of **Quiz Craft AI**, an intelligent MCQ quiz generation platform designed to improve assessment efficiency, enhance learning outcomes, and support modern educational environments through AI-powered automation.

1.6 Existing System

Currently, most educational institutions, coaching centers, and online learning platforms rely on traditional quiz systems for conducting assessments. These systems generally require teachers and instructors to manually create question banks, prepare quizzes, and manage assessments.

Existing Workflow

The instructor prepares study material and question banks manually.

Questions are selected and organized for quiz creation.

The quiz is created and published for students.

Students attempt the quiz.

The system evaluates responses and displays results.

Instructors review student performance and maintain records.

1. Limitations of Existing Systems

- Quiz creation requires significant manual effort and time.
- Most systems use fixed question patterns and predefined question banks.
- The same set of questions is often presented to all students.
- Increased chances of answer sharing and cheating during online assessments.
- Limited use of Artificial Intelligence for automatic question generation.
- Difficulty in creating large numbers of quality MCQs efficiently.
- Lack of dynamic and personalized quiz generation.
- Managing and updating question banks can be time-consuming.
- Repetitive assessments may reduce student engagement and learning effectiveness.
- Existing systems focus mainly on quiz delivery rather than intelligent content generation.

As a result, educators spend considerable time preparing assessments, while students often experience repetitive and predictable quizzes. These limitations highlight the need for a more intelligent and automated assessment system capable of generating high-quality MCQs dynamically and improving the overall learning experience.

1.7 Proposed System

proposed system, Quiz Craft AI, is an AI-powered MCQ Quiz Generator designed to automate quiz creation and improve the effectiveness of online assessments. The system utilizes Artificial Intelligence to generate multiple-choice questions.

Working of the Proposed System

1. The user uploads educational content, notes, or PDF documents.
2. The backend processes the uploaded content and extracts important information.
3. Artificial Intelligence analyzes the content and identifies key concepts and topics.
4. The system automatically generates multiple-choice questions based on the extracted information.
5. Generated questions are stored in the database for future use.
6. The system creates quizzes dynamically from the generated question bank.
7. Questions and answer options are shuffled automatically to ensure fairness.
8. Students attempt the generated quizzes through the web interface.
9. The system evaluates responses and calculates scores automatically.
10. Quiz results and performance records are stored in the database for future reference.

Benefits of the Proposed System

- Automatic generation of MCQs from educational content.
- Significant reduction in manual quiz preparation effort.
- Faster and more efficient assessment creation.
- Enhanced student engagement and self-learning.
- Instant evaluation and score calculation.
- Efficient management of question banks and quiz records.
- User-friendly interface for both students and educators.
- Better utilization of Artificial Intelligence in education.

The proposed system provides a smart, reliable, and efficient solution for modern educational environment.

1.8 Unique Features of the System

Quiz Craft AI incorporates several innovative features that distinguish it from traditional quiz management and assessment systems.

AI-Powered MCQ Generation

The system utilizes Artificial Intelligence to automatically generate multiple-choice questions from educational content, study materials, and PDF documents, significantly reducing manual effort.

Intelligent Content Analysis

The platform analyzes uploaded content, identifies key concepts, and generates relevant questions based on the extracted information, ensuring meaningful and topic-focused assessments.

Automatic Question Randomization

Questions and answer options are shuffled automatically for every quiz attempt, reducing predictability and minimizing the chances of cheating during online assessments.

Dynamic Quiz Creation

Unlike traditional systems that rely on fixed question banks, Quiz Craft AI generates quizzes dynamically, providing unique assessment experiences for different users.

Instant Evaluation and Scoring

The system automatically evaluates quiz responses and generates scores instantly, eliminating the need for manual assessment and improving efficiency.

Efficient Question Bank Management

Generated questions are stored systematically in a database, allowing efficient organization, retrieval, and reuse of educational content.

Self-Assessment and Practice Support

Students can generate practice quizzes from their study materials, helping them evaluate their knowledge and improve learning outcomes through continuous self-assessment.

AI-Assisted Educational Technology

The project demonstrates the practical implementation of Artificial Intelligence in education by automating quiz generation and enhancing digital assessment processes.

CHAPTER 2

REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

2.1 Feasibility Study

A feasibility study was conducted before the development of **Quiz Craft AI** to evaluate whether the project could be implemented successfully within the available resources, budget, and timeframe. The study focused on technical, economic, and operational feasibility.

2.1.1 Technical Feasibility

Technical feasibility examines whether the required technology, hardware, software, and expertise are available for successful project implementation.

The **Quiz Craft AI** system is technically feasible because it is developed using modern and widely adopted web technologies. The frontend is built using **HTML, CSS, and JavaScript**, which provide a responsive and interactive user interface. The backend utilizes **Python FastAPI** for handling API requests, business logic, and communication with AI services. **MongoDB** serves as the database for storing generated questions, quiz records, user information, and assessment results.

The integration of Artificial Intelligence enables the system to analyze educational content and automatically generate multiple-choice questions. The required development tools such as **Visual Studio Code, GitHub, MongoDB Compass, and Postman** are readily available and well-supported.

Since all technologies are stable, documented, and supported by large developer communities, the project can be implemented efficiently without significant technical barriers.

Technical Feasibility Conclusion

The project is technically feasible because all required technologies, tools, and resources are readily available and compatible with the proposed system architecture

2.1.2 Economic Feasibility

Economic feasibility determines whether the project can be developed and maintained within an acceptable budget.

The development cost of **Quiz Craft AI** is relatively low because most technologies used are open-source and freely available. HTML, CSS, JavaScript, Python FastAPI, MongoDB Community Edition, Git, and Visual Studio Code do not require licensing fees. Cloud deployment services such as Render, Vercel, Railway, or MongoDB Atlas offer free-tier plans that are sufficient for educational and prototype deployments. The only potential operational cost may arise from the usage of AI APIs if the system scales significantly and serves a large number of users.

The project does not require expensive hardware because development, testing, and deployment can be performed on a standard personal computer with an internet connection.

Economic Feasibility Conclusion

The overall development and deployment costs are minimal, making the project economically feasible for academic and practical implementation.

2.1.3 Operational Feasibility

Operational feasibility evaluates whether users will be able to effectively use the system after deployment.

The **Quiz Craft AI** platform is designed with simplicity and ease of use as primary objectives. Users only need to upload educational content or study material, and the system automatically generates multiple-choice questions and quizzes.

The interface presents information in a structured and understandable format using:

- AI-generated MCQs
- Automated quiz creation
- Randomized question arrangement
- Instant score calculation
- Quiz history and performance records
- User-friendly dashboards and reports

The system reduces the complexity of manual quiz preparation and helps both educators and students perform assessments more efficiently. Teachers can generate quizzes quickly, while students can evaluate their knowledge through practice assessments without requiring extensive technical knowledge.

Operational Feasibility Conclusion

The system is operationally feasible because it is intuitive, easy to learn, and capable of providing efficient quiz generation and assessment management with minimal training.

2.2 Software Requirement Specification (SRS)

The Software Requirement Specification defines the functional and non-functional requirements necessary for the successful operation of the **Quiz Craft AI** system

The system requires various categories of data for processing, storage, and quiz generation.

Input Data

The system accepts:

- Educational content in PDF, DOC, and text formats.
- User registration and authentication information.
- Study materials uploaded by users.
- Quiz configuration details such as quiz title, number of questions, and difficulty level.
- User responses submitted during quiz attempts.

Processed Data

After analysis, the system generates:

- Extracted educational content.
- Important topics and concepts.
- AI-generated multiple-choice questions.
- Answer options and correct answers.
- Quiz scores and performance results.
- Assessment reports and quiz statistics.

Stored Data

The database stores:

- User profiles.
- Uploaded study materials.
- Generated MCQs.
- Quiz records and results.
- Performance analytics and assessment history.

2.2.2 Functional Requirements

Functional requirements define what the system must do.

User Authentication Module

The system shall:

- Allow user registration.
- Allow secure login and logout.
- Maintain authenticated user sessions.

Content Upload Module

The system shall:

- Accept PDF and educational documents.
- Validate file type and size.
- Upload content securely to the server.

AI Quiz Generation Module

The system shall:

- Process uploaded study materials.
- Extract important educational information.

- Generate multiple-choice questions automatically.
- Create answer options and identify correct answers.

Quiz Management Module

The system shall:

- Create quizzes dynamically.
- Shuffle questions and answer options.
- Manage quiz attempts.
- Store quiz-related information.

Evaluation Module

The system shall:

- Evaluate user responses automatically.
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Dashboard Module

The system shall:

- Display quiz history.
- Show performance statistics.
- Visualize assessment results.
- Present learning progress information.

History Module

The system shall:

- Store previous quiz attempts.
- Allow retrieval of past assessments.
- Maintain long-term performance records.

2.2.3 Performance Requirements

The system should provide fast and reliable performance.

Response Time

- File uploads should complete within a few seconds.
- AI-based quiz generation should be completed within an acceptable response time.
- Quiz results should be displayed instantly after submission.
- Dashboard data should load quickly.

Scalability

The system should support:

- Multiple users simultaneously.
- Large numbers of generated quizzes.
- Large educational content repositories.
- Future expansion of features and services.

Availability

The application should remain accessible and functional with minimal downtime.

Reliability

The system should consistently generate accurate MCQs and handle unexpected errors gracefully.

2.2.4 Dependability Requirements

Dependability refers to the reliability and trustworthiness of the system.

The system should:

- Produce consistent results.
- Prevent accidental data loss.
- Handle API failures properly.
- Recover gracefully from system errors.
- Maintain data integrity during storage and retrieval operations.

Backup mechanisms and error handling procedures help improve system dependability.

2.2.5 Maintainability Requirements

Maintainability ensures that the system can be modified and improved over time.

The Quiz Craft AI architecture follows a modular design where frontend, backend, AI processing, and database layers are separated.

This enables:

- Easier debugging.
- Faster feature additions.
- Simpler code maintenance.
- Improved scalability.
- Better system flexibility.

Version control using Git ensures effective change management and collaborative development.

2.2.6 Security Requirements

Security is important because the system processes user information, educational content, generated quizzes, and assessment records.

The system implements the following security measures:

Authentication Security

- Secure user registration and login system.
- Password encryption before storage.
- Session management and user authentication.

API Security

- Environment variables for storing API keys securely.
- Restricted API access.
- Input validation and request verification.

Data Security

- Secure storage of user profiles and quiz records.
- Controlled access to educational content and generated questions.
- Protection against unauthorized modifications and data breaches.

Application Security

- Prevention of malicious file uploads.
- Input sanitization to prevent security vulnerabilities.
- Cross-Origin Resource Sharing (CORS) protection.
- Secure communication between frontend and backend services.

These security measures help maintain the confidentiality, integrity, and availability of system data.

2.2.7 Look and Feel Requirements

The user interface should provide a professional, attractive, and intuitive experience for both students and educators.

The system shall:

- Use a responsive design.
- Support desktop, tablet, and mobile devices.
- Display clear navigation menus and user-friendly controls.
- Provide a clean and organized layout.
- Maintain visual consistency across all pages.
- Present quizzes and results in an easy-to-understand format.
- Ensure accessibility and readability for all users.
- Provide smooth interaction and navigation throughout the platform.

The interface follows a modern educational design philosophy with emphasis on simplicity, usability, readability, and user engagement. The design aims to provide a seamless experience while making quiz generation, quiz participation, and performance evaluation efficient and enjoyable for users

2.3 Validation

Validation ensures that the developed system satisfies the specified requirements

Validation activities included:

Functional Validation

Verification that all modules perform their intended functions correctly.

Data Validation

Verification that uploaded educational content is correctly processed, analyzed, and stored in the database.

AI Validation

Verification that the generated multiple-choice questions are relevant, accurate, and aligned with the uploaded study material.

Quiz Validation

Verification that quiz generation, question randomization, score calculation, and result generation function correctly.

Verification that all modules work together as a complete integrated system.

The validation process confirmed that the system fulfills its intended objectives and provides an efficient platform for AI-powered quiz generation and assessment

2.4 Expected Hurdles

Several challenges were anticipated during development.

Educational Content Variability

Different study materials, notes, and PDF documents may have different formats and structures, making content extraction challenging.

AI Question Generation Accuracy

Generating meaningful, accurate, and non-repetitive MCQs requires proper AI processing and validation.

API Limitations

Dependence on external AI services may introduce rate limits, response delays, or temporary service disruptions.

Content Processing Complexity

Extracting important concepts from lengthy educational documents can sometimes affect question quality and relevance.

Data Security Concerns

Handling user information, uploaded documents, and quiz records requires proper security measures and access control.

Scalability Challenges

As the number of users grows, increased database storage and processing requirements may arise.

CHAPTER 3

SYSTEM DESIGN

Object-Oriented Design Approach

Quiz Craft AI follows an Object-Oriented Design (OOD) approach. The system is divided into independent modules such as User Management, Study Material Management, AI-Based MCQ Generation, Quiz Management, Result Evaluation, Performance Analytics, and History Management. Each module is designed as a separate component that performs a specific task while interacting with other modules whenever required. This approach improves the overall organization of the system and makes development, testing, and maintenance more efficient.

Object-Oriented Design was selected because it provides a structured way to model real-world entities and their interactions. In the Quiz Craft AI system, entities such as User, StudyMaterial, AIQuestionGenerator, Quiz, Question, Result, Dashboard, and History are represented as classes. These classes encapsulate both data and functionality, making the system easier to understand, maintain, and extend.

The object-oriented approach improves:

•Modularity

The system is divided into multiple independent modules. Each module performs a specific responsibility such as content processing, question generation, quiz management, or result evaluation, reducing complexity and improving system organization.

•Reusability

Classes and components can be reused across different parts of the application. This minimizes code duplication and allows faster development of new features.

•Maintainability

Updates and modifications can be performed on individual modules without affecting the complete system. This simplifies debugging, testing, and future maintenance.

•Scalability

The architecture supports future enhancements such as adaptive learning, personalized quiz recommendations, AI-powered difficulty adjustment, advanced analytics, and support for multiple content formats.

Major Classes

User

Stores user authentication details and manages interactions with the system. It maintains user profiles, login credentials, uploaded materials, and quiz participation records.

StudyMaterial

Represents uploaded educational resources such as PDF files, notes, documents, and text content. It manages storage, processing status, and extracted learning information.

AIQuestionGenerator

The core class responsible for analyzing study material and generating intelligent multiple-choice questions. It processes educational content and creates relevant questions along with answer options.

Quiz

Represents generated quizzes and maintains quiz-related information such as quiz title, generated questions, duration, difficulty level, and quiz settings.

Question

Stores AI-generated MCQs, answer options, correct answers, explanations, topic information, and difficulty levels.

Result

Maintains quiz scores, performance summaries, accuracy percentages, and assessment outcomes generated after quiz completion.

Dashboard

Provides users with quiz statistics, performance analytics, progress tracking, and access to previously attempted quizzes.

History

Stores records of previous quiz attempts, generated quizzes, scores, and historical performance data for future reference and learning analysis.

The Object-Oriented Design approach enables Quiz Craft AI to provide a reliable, intelligent, scalable, and maintainable educational platform while ensuring efficient management of study materials, AI-generated quizzes, user interactions, and performance evaluation.

3.1 Design Approach

The design approach of Quiz Craft AI – AI Driven MCQ Quiz Generator focuses on developing a simple, efficient, and intelligent educational platform that automates the process of quiz creation from study materials. The system is designed using a modular

and object-oriented architecture, where each component performs a specific task while interacting with other components to achieve the overall functionality of the application. The development of the system follows a layered architecture consisting of the Presentation Layer, Application Layer, AI Processing Layer, and Data Layer. This structure helps in separating user interface, business logic, AI-based question generation, and data management activities. Such separation improves system maintainability, scalability, and performance.

The design begins with the Study Material Management module, where users can upload educational content in the form of PDF documents, notes, or textual material. The uploaded content is processed and analyzed to extract important concepts, keywords, and learning topics. This information serves as the input for the AI-based question generation process.

The core component of the system is the AI Question Generation module. This module analyzes the uploaded content and automatically generates meaningful multiple-choice questions along with answer options and correct answers. The generated questions are stored in the database and can be used to create quizzes instantly. This automation significantly reduces manual effort and improves the efficiency of quiz preparation.

The Quiz Management module is responsible for organizing generated questions into quizzes. It manages quiz settings, question selection, difficulty levels, and quiz execution. To ensure fairness and prevent copying, the system applies randomization techniques that shuffle questions and answer options whenever a quiz is attempted.

The Result Evaluation module evaluates user responses after quiz submission. It compares submitted answers with correct answers stored in the system and calculates scores automatically. The generated results are then presented to users in an easy-to-understand format.

To enhance the learning experience, the Dashboard and Analytics module provides performance reports, quiz statistics, and historical records of previous quiz attempts. This helps users track their progress and identify areas that require improvement.

The database design stores user information, uploaded study materials, generated questions, quizzes, and result records in a structured manner. Proper data organization ensures efficient retrieval and management of information.

Overall, the design approach of Quiz Craft AI emphasizes automation, intelligence, scalability, and user-friendliness. The modular architecture allows future enhancements such as adaptive learning, personalized recommendations, advanced AI models, and detailed performance analytics without major modifications to the existing system.

3.2 Detailed Design

3.2.1 System Architecture

4

Quiz Craft AI follows a three-tier architecture consisting of the Presentation Layer, Business Logic Layer, and Data Layer. This architecture separates the user interface, application processing, AI-powered quiz generation, and database management into independent layers, improving maintainability, scalability, and system performance. The layered design allows efficient communication between different system components while ensuring that each layer performs a specific responsibility.

Presentation Layer

The Presentation Layer serves as the user interface of the Quiz Craft AI system. It enables users to interact with the application and access various features such as uploading study materials, generating AI-based quizzes, attempting assessments, viewing quiz results, and tracking performance.

Components used in the Presentation Layer include:

HTML, CSS, and JavaScript Frontend

HTML, CSS, and JavaScript are used to develop a responsive and interactive user interface. These technologies provide a user-friendly environment where users can upload PDF documents, generate quizzes, attempt MCQs, and view results efficiently.

Responsive User Interface

The interface is designed to function smoothly across multiple devices, including desktops, laptops, tablets, and smartphones. The design emphasizes simplicity, accessibility, and ease of navigation to improve the user experience.

Business Logic Layer

The Business Logic Layer acts as the core processing unit of the Quiz Craft AI system. It receives requests from the frontend, processes uploaded study materials, communicates with AI services, generates MCQs, manages quizzes, and handles result evaluation.

Components used in the Business Logic Layer include:

Python FastAPI

FastAPI serves as the backend framework responsible for handling API requests, business logic, user interactions, quiz generation, and communication with the database. It provides high performance and efficient request processing.

AI MCQ Generation Module

The AI module is the most important component of the system. It analyzes uploaded study materials such as PDF files, notes, and textual content. Using Artificial Intelligence techniques, the module extracts important concepts and automatically generates multiple-choice questions along with answer options and correct answers.

Quiz Management Module

This module manages quiz creation, question selection, question randomization, answer validation, score calculation, and result generation. It ensures fair assessment by providing randomized quizzes for users.

Result Evaluation Module

The result evaluation module compares user responses with correct answers and calculates scores automatically. It also generates performance summaries and assessment reports.

Data Layer

The Data Layer is responsible for storing, retrieving, and managing application data. It maintains user information, uploaded study materials, AI-generated questions, quiz records, results, and historical performance data.

Components used in the Data Layer include:

MongoDB Database

MongoDB is used as the primary NoSQL database for storing application data. Its flexible document-based structure allows efficient storage and retrieval of study materials, generated MCQs, quiz information, user data, and assessment results.

System Workflow

The overall workflow of the Quiz Craft AI system is illustrated below:

User → Frontend Interface → FastAPI Backend → AI MCQ Generation Module → MongoDB Database → Quiz Interface → Result Evaluation → Dashboard → User

The user uploads study material through the frontend interface. The FastAPI backend receives the uploaded content and sends it to the AI MCQ Generation Module for processing. The AI module analyzes the content and generates relevant multiple-choice questions. These generated questions are stored in the MongoDB database and displayed to the user as a quiz. After quiz completion, the system evaluates responses, calculates scores, stores results, and presents performance reports through the dashboard.

Advantages of the Three-Tier Architecture

- Improved Scalability
- Better Maintainability
- Enhanced Security
- Efficient Data Management
- Faster Quiz Generation
- Easier Integration of AI Services

- Separation of Concerns
- Improved User Experience
- Faster Development and Testing
- Better System Performance

3.3 System Design Using UML and Structured Analysis Tools

3.3.1 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of the movement of data within a system. It illustrates how data is collected, processed, stored, and transferred between different components. The DFD helps in understanding the overall working of the Quiz Craft AI system by showing the flow of information from study material upload to AI-based MCQ generation, quiz execution, and result evaluation.

The Level 0 DFD represents the overall flow of data in the Quiz Craft AI system. The user uploads educational content or study material, which is processed by the system using Artificial Intelligence techniques. The generated multiple-choice questions are then stored in the database and displayed to the user as a quiz for assessment.

The DFD begins when a user uploads study material such as PDF documents, notes, textbooks, or educational content through the application interface. The uploaded content is forwarded to the content processing module, where important concepts, topics, and keywords are extracted from the document. The extracted information is then analyzed by the AI MCQ Generation Engine.

Based on the content analysis, the system automatically generates multiple-choice questions along with answer options and correct answers. The generated questions are organized into a quiz and stored in the MongoDB database. The quiz is then displayed to the user through the frontend interface.

After the user attempts and submits the quiz, the Result Evaluation Module processes the submitted answers and compares them with the correct answers stored in the database. The system calculates the final score, accuracy percentage, and overall performance summary.

The generated results are stored in the database and displayed through the user dashboard. Users can view their quiz history, previous scores, performance reports, and assessment records for future reference and self-improvement.

Thus, the DFD demonstrates the complete flow of data across the Quiz Craft AI platform, starting from study material upload and AI-powered MCQ generation to quiz participation, result evaluation, and performance tracking.

Advantages of DFD

- Provides a clear understanding of data movement.
- Simplifies system analysis and design.
- Helps identify system inputs and outputs.
- Improves communication among developers and stakeholders.
- Assists in identifying potential bottlenecks and improvements.
- Provides better visualization of system processes.
- Helps understand AI-based content processing and quiz generation workflows.
- Supports efficient planning and implementation of system modules.

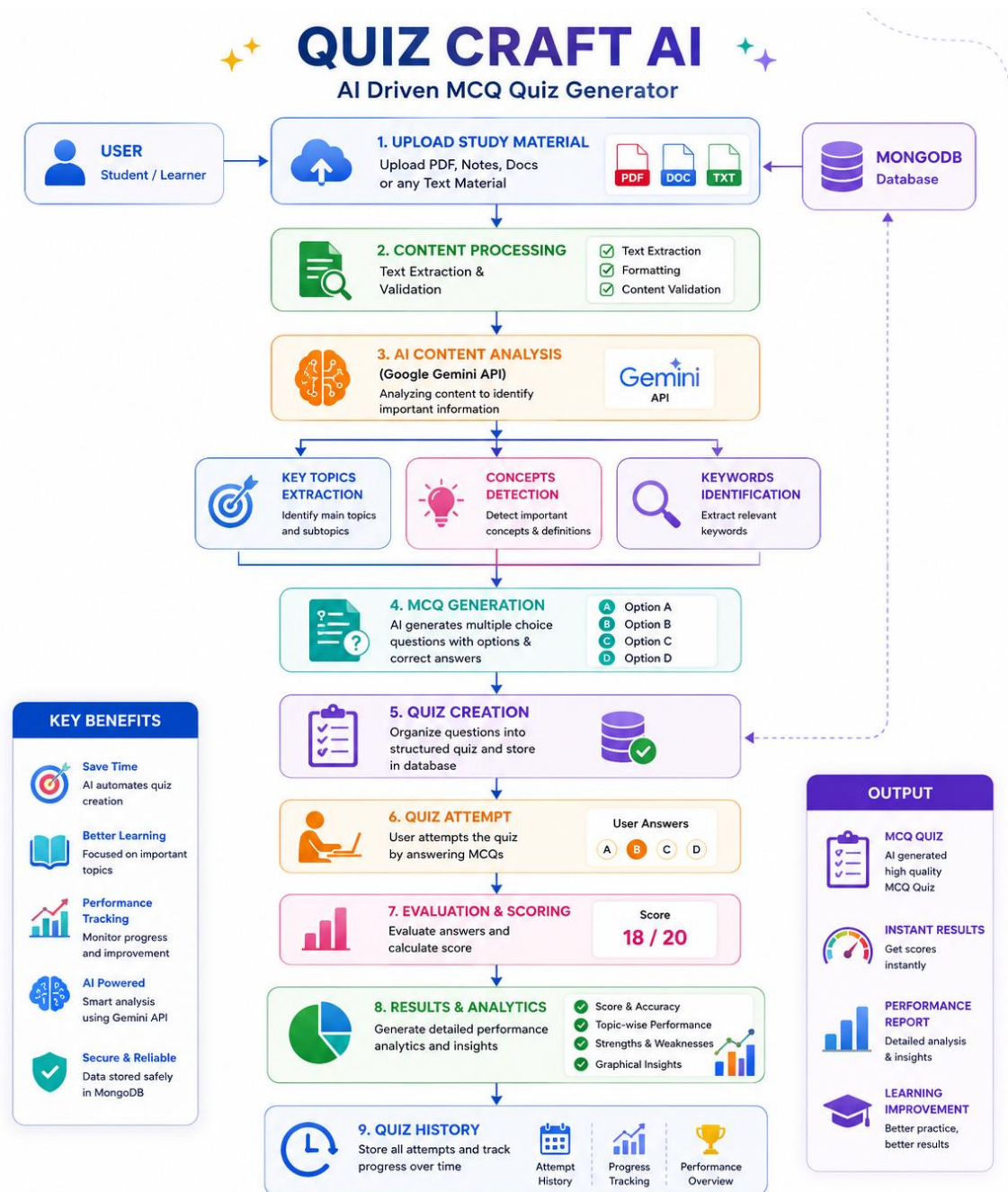


Figure 3.1: Level 0 Data Flow Diagram of AI QUIZ Craft

3.3.2 Use Case Diagram

A Use Case Diagram describes the interactions between external actors and the Quiz Craft AI system. It identifies the major functionalities provided by the system and the users who interact with them. The diagram helps in understanding system requirements from the user's perspective.

The Use Case Diagram represents the interaction between different actors and the Quiz Craft AI system. It illustrates the major functionalities available to users and the actions performed within the application

Actors

- **Student/User**

The primary actor who registers into the system, uploads study materials, generates AI-based MCQs, attempts quizzes, views results, and tracks learning progress through the dashboard. The user can also access previous quiz records and performance reports.

- **Admin**

Responsible for managing users, monitoring system activities, maintaining uploaded study materials, managing quiz records, and ensuring the smooth operation of the platform. The admin can also monitor generated quizzes and maintain database records.

- **AI MCQ Generation Engine**

Acts as an intelligent system component that analyzes uploaded educational content, extracts important concepts, generates multiple-choice questions, creates answer options, and supports automated quiz creation

Use Cases

- Register/Login
- Upload Study Material
- Process Educational Content
- Generate AI-Based MCQs

- Create Quiz
- Attempt Quiz
- Submit Quiz
- Evaluate Responses
- View Results
- View Dashboard
- View Performance Reports
- View Quiz History
- Manage Profile
- Manage Users (Admin)
- Manage Quiz Records (Admin)
- Monitor System Activity (Admin)

Description

The Use Case Diagram provides a high-level overview of the services offered by Quiz Craft AI and highlights the interaction between users and system functionalities. The user uploads study materials such as PDF files, notes, or textual content. The AI MCQ Generation Engine analyzes the uploaded content and automatically generates relevant multiple-choice questions. Users can attempt generated quizzes, view their scores, and track performance through the dashboard.

The Admin supervises overall system activities, manages users, and maintains quiz-related information. The AI engine works as the core component that automates question generation and reduces the manual effort required for quiz creation.

This diagram helps developers and stakeholders understand how different actors interact with the platform and utilize its AI-powered quiz generation and assessment capabilities.

Advantages of Use Case Diagram

- Provides a clear understanding of system functionality.
- Identifies user interactions with the system.
- Helps gather and validate system requirements.
- Improves communication between developers and stakeholders.

- Assists in system design and development planning.
- Simplifies understanding of user roles and responsibilities.
- Clearly represents AI-powered quiz generation workflows.
- Helps visualize system services and user activities.

3.3.3 Class Diagram

The Class Diagram represents the static structure of the Quiz Craft AI system. It illustrates the various classes, their attributes, methods, and relationships. The diagram helps developers understand how different components interact within the application and provides a blueprint for implementing the system using object-oriented principles.

Classe

• User

Stores user information such as name, email, password, login credentials, profile details, and authentication information. It manages user activities including study material uploads, quiz participation, and performance tracking.

• StudyMaterial

Represents uploaded educational resources such as PDF documents, notes, textbooks, and learning materials. It stores information such as file name, upload date, content details, and processing status.

AIQuestionGenerator

Represents the AI-powered module responsible for analyzing uploaded study materials and generating multiple-choice questions. It extracts important concepts, identifies key topics, creates answer options, and generates correct answers automatically.

Quiz

Represents generated quizzes and stores quiz-related information such as quiz title, creation date, total number of questions, difficulty level, and quiz settings.

Question

Stores AI-generated multiple-choice questions, answer options, correct answers, explanations, and topic information associated with a particular quiz.

Result

Represents quiz outcomes and stores information such as obtained score, total marks, percentage, accuracy level, completion status, and assessment summaries.

Dashboard

Provides a visual representation of quiz performance, learning progress, assessment statistics, generated quiz summaries, and historical records.

History

Maintains records of previous quiz attempts, generated quizzes, user activities, and assessment data for future reference and progress analysis.

Relationships

- User uploads multiple Study Materials.
- Study Material is processed by the AIQuestionGenerator.
- AIQuestionGenerator generates multiple Questions.
- Questions are organized into a Quiz.
- User attempts multiple Quizzes.
- Quiz contains multiple Questions.
- Quiz generates Results after evaluation.
- Results are displayed on the Dashboard.
- User can access History records.
- Dashboard retrieves information from Results and History.

Description

The Class Diagram helps in implementing object-oriented design principles and provides a clear representation of system entities, their responsibilities, and relationships. Each class performs a specific function within the Quiz Craft AI platform, ensuring modularity, maintainability, and scalability.

The User class interacts with the system by uploading study materials and attempting quizzes. The StudyMaterial class stores educational content, which is processed by the AIQuestionGenerator class. The generated questions are organized into quizzes through the Quiz class. After quiz completion, the Result class stores assessment outcomes, while the Dashboard and History classes provide performance insights and historical records.

Advantages of Class Diagram

- Provides a clear view of system structure.
- Supports object-oriented design principles.
- Improves software maintainability.
- Helps developers understand class relationships.
- Simplifies implementation and future enhancements.
- Provides a foundation for database and application design.
- Enhances system modularity and reusability.
- Supports scalable AI-based application development.

QUIZ CRAFT AI – AI DRIVEN MCQ QUIZ GENERATOR

CLASS DIAGRAM

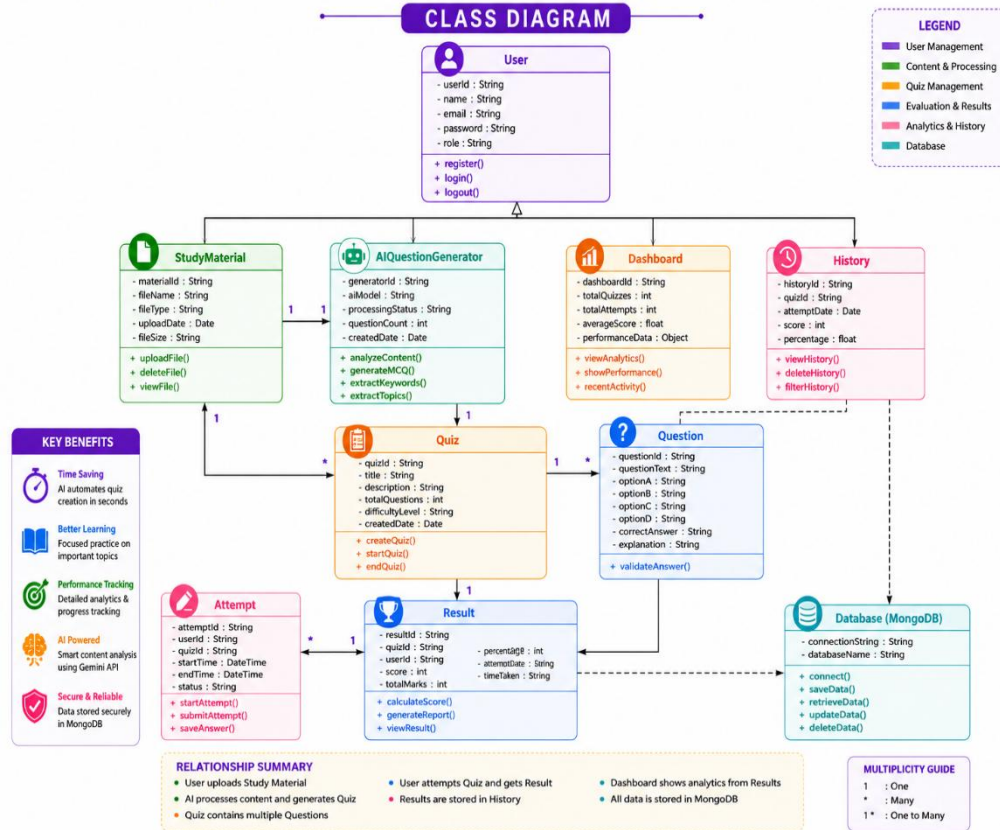


Figure 3.3: Class Diagram of Quiz Craft AI

3.3.4 Sequence Diagram

A Sequence Diagram illustrates the chronological order of interactions among different system components. It shows how requests are processed and how messages are exchanged between the frontend, backend, AI modules, and database components of the Quiz Craft AI system.

The sequence begins when a user uploads study material such as PDF documents, notes, or educational content through the frontend interface. The frontend sends the uploaded content to the FastAPI backend server for validation and processing. The backend verifies the uploaded file, extracts relevant content, and forwards the processed data to the AI MCQ Generation Module.

The AI module analyzes the educational content, identifies important concepts and topics, and automatically generates multiple-choice questions along with answer options and correct answers. The generated quiz data is then returned to the backend server.

The backend stores the generated questions, quiz information, and related metadata in the MongoDB database. Once the quiz is successfully stored, the backend sends the generated quiz information back to the frontend interface for display.

The user can then attempt the generated quiz through the web application. After answering all questions and submitting the quiz, the frontend sends the user responses to the backend for evaluation.

The backend compares the submitted answers with the correct answers stored in the database. It calculates the score, accuracy percentage, and overall performance summary. The assessment results are then stored in MongoDB for future reference.

Finally, the dashboard retrieves the stored results and displays quiz scores, performance statistics, progress reports, and quiz history to the user. Users can review previous assessments and monitor their learning progress over time.

Thus, the Sequence Diagram demonstrates the complete interaction flow of the Quiz Craft AI system, from study material upload and AI-powered question generation to quiz evaluation and result visualization.

Benefits of Sequence Diagram

- Shows the order of system operations.
- Helps identify interactions between components.
- Simplifies understanding of system workflow.
- Assists in debugging and testing.
- Provides better visualization of message flow.
- Improves system design documentation.
- Helps understand AI-powered quiz generation processes.
- Clarifies communication between frontend, backend, AI modules, and database.

Steps

1. User uploads study material.
2. Frontend sends uploaded content to the FastAPI backend.
3. Backend validates and processes the content.
4. Backend forwards content to the AI MCQ Generation Module.
5. AI module analyzes content and extracts key concepts.
6. AI generates MCQs, answer options, and correct answers.
7. Backend stores generated quiz data in MongoDB.
8. Generated quiz is displayed to the user.
9. User attempts and submits the quiz.
10. Backend evaluates responses and calculates the score.
11. Results are stored in MongoDB.
12. Dashboard retrieves and displays results.
13. User views performance reports and quiz history.
14. Conclusion

The Sequence Diagram provides a detailed view of how different components interact within the Quiz Craft AI platform. It helps developers understand the flow of messages, system behavior, and processing sequence involved in AI-based quiz generation, assessment management, and performance evaluation.

Steps

1. User uploads study material.
2. Frontend sends content to backend.
3. Backend validates and processes content.
4. AI Question Generation Module analyzes content.
5. AI generates MCQs and quiz data.
6. Backend stores generated quiz in MongoDB.
7. Quiz is displayed to the user.
8. User attempts and submits quiz.
9. Backend evaluates responses and calculates score.
10. Dashboard updates and displays results.

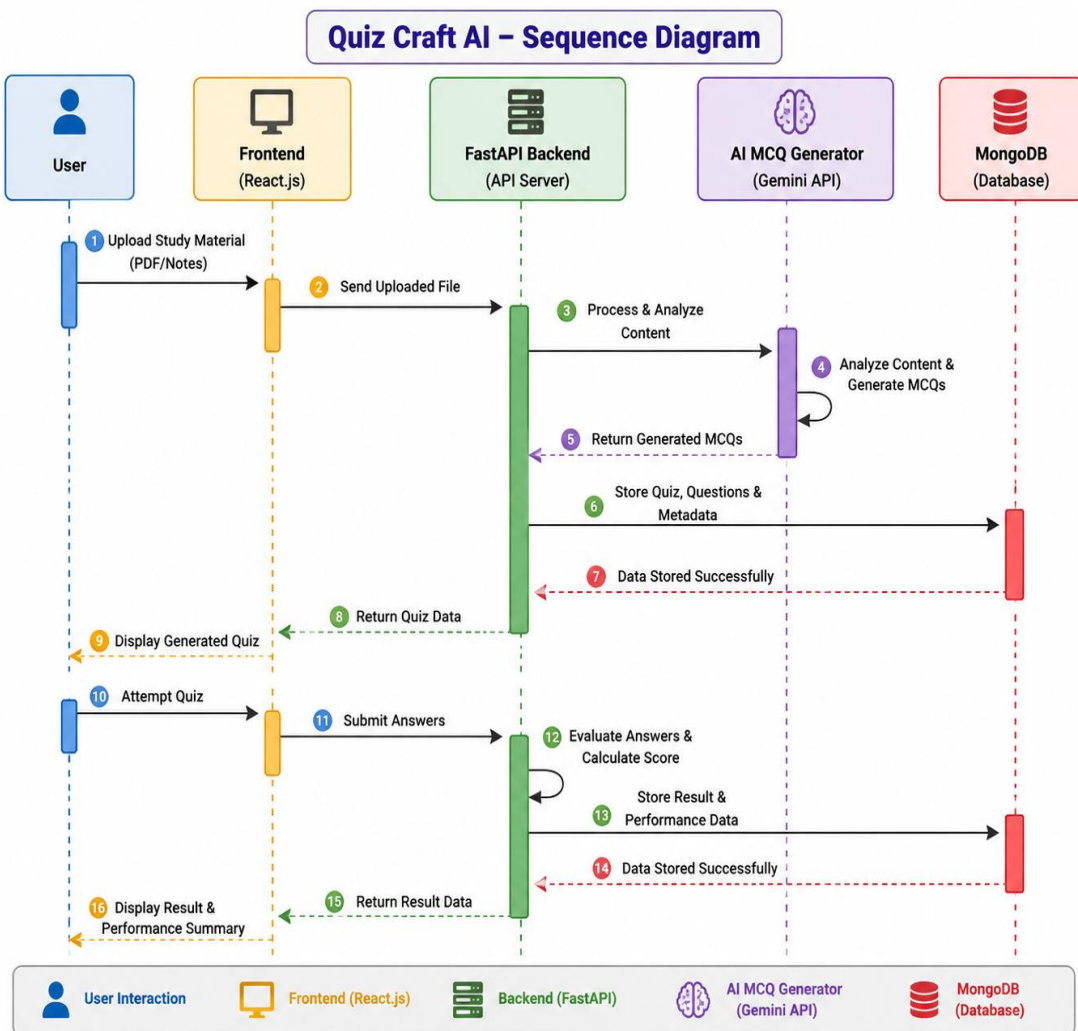


Figure 3.4: Sequence Diagram of Quiz Craft AI

3.3.5 Flowchart

A flowchart is a graphical representation of the sequence of activities performed within a system. It helps visualize the logical flow of operations, processing steps, and decision-making activities involved in the execution of the Quiz Craft AI system.

The Quiz Craft AI flowchart begins when a user accesses the application and performs authentication through login or registration. After successful authentication, the user uploads study material such as PDF documents, notes, textbooks, or educational content through the system interface. The uploaded content is then validated to ensure that it is in a supported format and contains readable information.

Once validation is completed successfully, the content is forwarded to the AI MCQ Generation Module for processing. The AI engine analyzes the educational content, extracts important concepts, keywords, and topics, and automatically generates multiple-choice questions along with answer options and correct answers.

The generated quiz is then stored in the MongoDB database and made available to the user through the quiz interface. The user attempts the generated quiz by selecting answers for each question and submits the quiz after completion.

After submission, the system evaluates the user responses by comparing them with the correct answers stored in the database. Based on the evaluation process, the system calculates the final score, accuracy percentage, and overall performance summary.

The generated results are stored in the database and displayed through the user dashboard. The dashboard provides quiz scores, performance reports, learning progress, and historical quiz records. Users can review previous quiz attempts and track their improvement over time.

The process concludes when the user views the generated results and performance statistics. Thus, the flowchart provides a complete representation of the workflow involved in AI-based quiz generation, quiz participation, result evaluation, and performance tracking.

Advantages of Flowcharts

- Simplifies understanding of system processes.
- Provides visual representation of workflow.
- Helps identify decision points.
- Assists in system documentation and maintenance.
- Improves communication among development team members.
- Facilitates debugging and process optimization.
- Helps visualize AI-powered quiz generation workflow.
- Improves understanding of system execution flow.

Flowchart Steps

Start



Login/Register



Upload Study Material (PDF/Notes/Text Content)



Validate Uploaded Content



Process Educational Content



AI Content Analysis



Generate AI-Based MCQs



Store Generated Quiz in MongoDB



Display Quiz to User



Attempt Quiz



Submit Responses



Evaluate Responses



Calculate Score & Performance



Store Results in Database



Display Dashboard & Quiz History



End

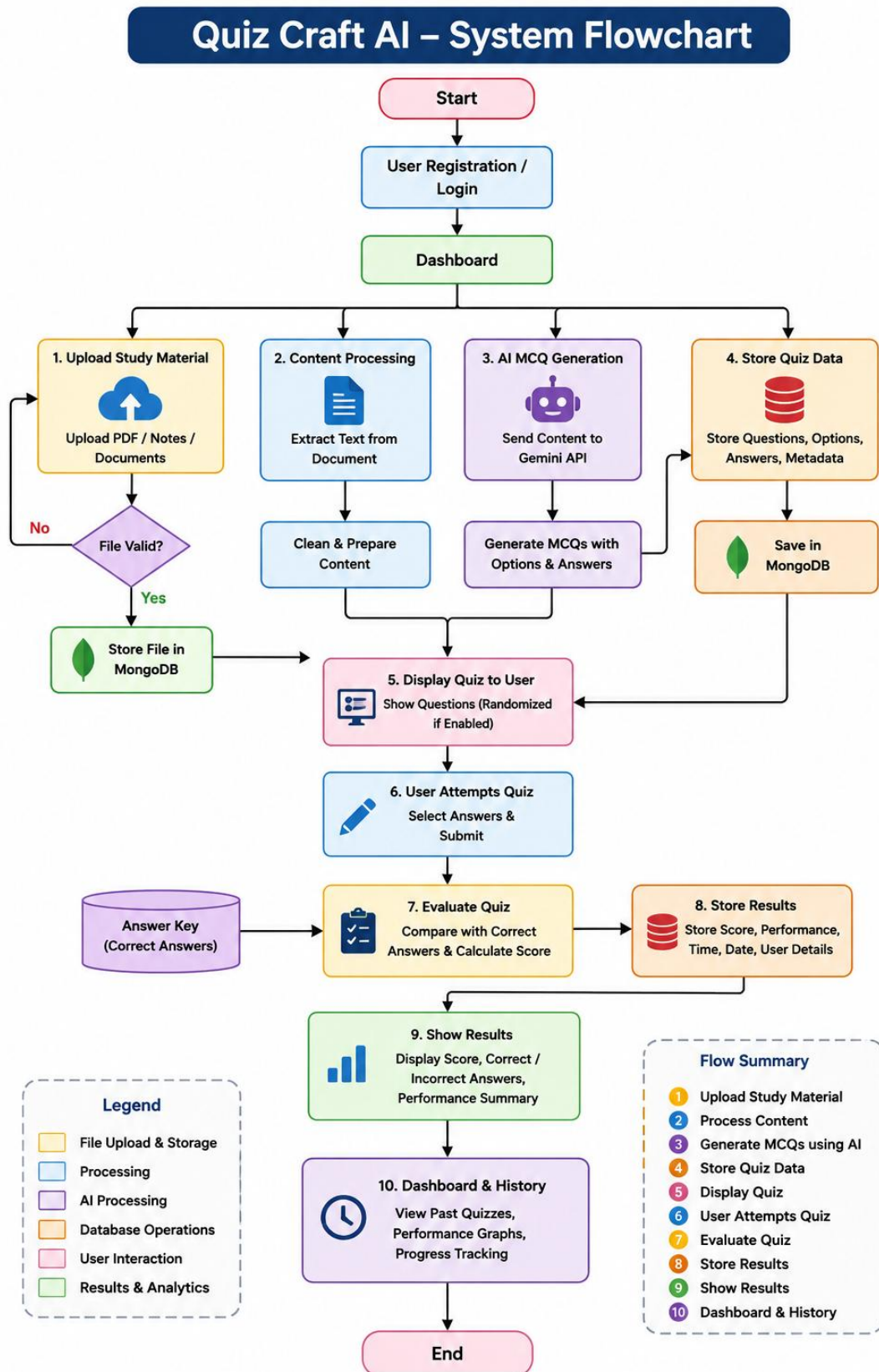


Figure 3.5: Flowchart of Quiz Craft AI

3.4 User Interface Design

The user interface of Quiz Craft AI is designed to provide a simple, responsive, and user-friendly experience for students and educators. The interface enables users to upload study materials, generate AI-powered quizzes, attempt assessments, view results, and track performance efficiently. The UI is developed using HTML, CSS, and JavaScript, ensuring a modern design and smooth user interaction across different devices.

3.4.1 Landing Page

The Landing Page serves as the entry point of the Quiz Craft AI system. It introduces users to the platform and provides an overview of the major features available within the application. The page is designed to be simple and informative, allowing users to quickly understand the purpose of the system.

Purpose

- Introduce Quiz Craft AI and its objectives.
- Display key features and benefits.
- Provide navigation to major modules.
- Guide users to quiz generation and assessment features.

Features

- Clean and responsive user interface.
- Easy navigation menu.
- Feature highlights and system overview.
- Quick access to login and registration.

3.4.2 Study Material Upload Interface

The Study Material Upload Interface enables users to upload educational content such as PDF documents, notes, and learning materials for AI-based quiz generation. The interface provides a simple and convenient upload mechanism for users.

Features

- Drag and Drop Upload functionality.
- PDF and document file support.
- File validation before submission.

- Upload progress indication.
- Secure content handling and storage.

The uploaded content is forwarded to the backend server for processing and AI-based MCQ generation.

3.4.3 Quiz Generation Interface

The Quiz Generation Interface displays AI-generated multiple-choice questions based on the uploaded study material. It presents questions in a structured and easy-to-understand format, enabling users to attempt quizzes efficiently.

Features

- AI-generated MCQs.
- Multiple answer options.
- Automatic question randomization.
- Topic-based question generation.
- User-friendly quiz presentation.
- Instant quiz creation from uploaded content.

This interface helps users quickly generate assessment-ready quizzes without manual effort.

3.4.4 Dashboard Interface

The Dashboard serves as the central information hub of the Quiz Craft AI system. It provides users with an overview of generated quizzes, assessment records, performance statistics, and learning progress.

Features

- Quiz performance visualization.
- Summary of recent quiz attempts.
- Score and progress tracking.
- Quick access to generated quizzes.
- Learning analytics and assessment statistics.
- Performance monitoring dashboard.

The dashboard allows users to monitor their learning progress and assessment performance over time.

3.4.5 History Interface

The History Interface maintains a record of previously generated quizzes and assessment results. It enables users to revisit earlier quizzes whenever required.

Features

- Search quizzes by keywords.
- Filter quizzes by topic or subject.
- View previous quiz attempts.
- Access historical assessment records.
- Organized quiz management.
- Track learning progress over time.

The history module improves accessibility and allows users to review previous learning activities and performance records.

3.4.6 Result Interface

The Result Interface displays quiz outcomes and performance summaries after quiz completion. It helps users evaluate their understanding of the subject matter and identify areas for improvement.

Features

- Instant score calculation.
- Performance summary display.
- Correct and incorrect answer analysis.
- Percentage and grade calculation.
- Learning improvement suggestions.
- Detailed assessment feedback.

This interface enables users to identify strengths and weaknesses and improve their learning effectiveness.

3.4.7 User Profile Interface

The User Profile Interface allows users to manage their account information, preferences, and learning activities through a centralized dashboard.

Features

- View and update profile information.

- Manage account settings.
- Access quiz history.
- Track learning progress.
- Secure account management.
- Personalized user experience.

The profile module enhances personalization and provides a centralized location for managing user-related information.

3.5 Database Design

3.5.1 ER Diagram

The Entity Relationship (ER) Diagram represents the database structure of the Quiz Craft AI system. It illustrates the entities involved in the application, their attributes, and the relationships between them. The ER Diagram helps in understanding how data is organized and stored within the MongoDB database.

The Quiz Craft AI database is designed to manage user information, uploaded study materials, AI-generated quizzes, questions, and assessment results. Each entity plays a specific role in maintaining the functionality of the system.

Entities

User

The User entity stores authentication and account-related information. It contains details such as user ID, name, email address, password, profile information, and account settings. Each user can upload study materials, generate quizzes, attempt assessments, and view performance reports.

StudyMaterial

The StudyMaterial entity stores information related to uploaded educational content. It contains details such as material ID, file name, upload date, file type, subject, and file location. These study materials serve as the primary input for AI-based MCQ generation.

Quiz

The Quiz entity stores information related to generated quizzes. It contains details such as quiz ID, quiz title, creation date, total number of questions, difficulty level, and associated study material.

Question

The Question entity stores AI-generated multiple-choice questions. It contains question text, answer options, correct answers, difficulty level, explanation, and topic information. Each question belongs to a specific quiz.

Result

The Result entity stores assessment outcomes generated after quiz completion. It includes score, percentage, completion date, accuracy level, and performance statistics. Each result is associated with a particular user and quiz.

History

The History entity maintains records of previous quiz attempts, generated quizzes, user activities, and assessment history. It helps users track learning progress and review past assessments for improvement.

Relationships

- A User can upload multiple Study Materials.
- A Study Material can generate multiple Quizzes.
- A Quiz contains multiple Questions.
- A User can attempt multiple Quizzes.
- Each Quiz generates one or more Result records.
- A User can have multiple History records.
- Results and quiz activities are stored in History for performance tracking.

The ER Diagram provides a clear representation of the database structure and helps ensure efficient data storage, retrieval, and management within the Quiz Craft AI system.

ER Diagram Structure

User → StudyMaterial → Quiz → Question

User → Quiz → Result

User → History

Result → History

Advantages of ER Diagram

- Provides a clear view of database entities and relationships.
- Helps in designing an efficient database structure.
- Improves data organization and consistency.
- Simplifies database implementation and maintenance.
- Supports efficient retrieval and storage of information.
- Provides a foundation for application development.
- Helps developers understand entity relationships.
- Facilitates future system enhancements and scalability.

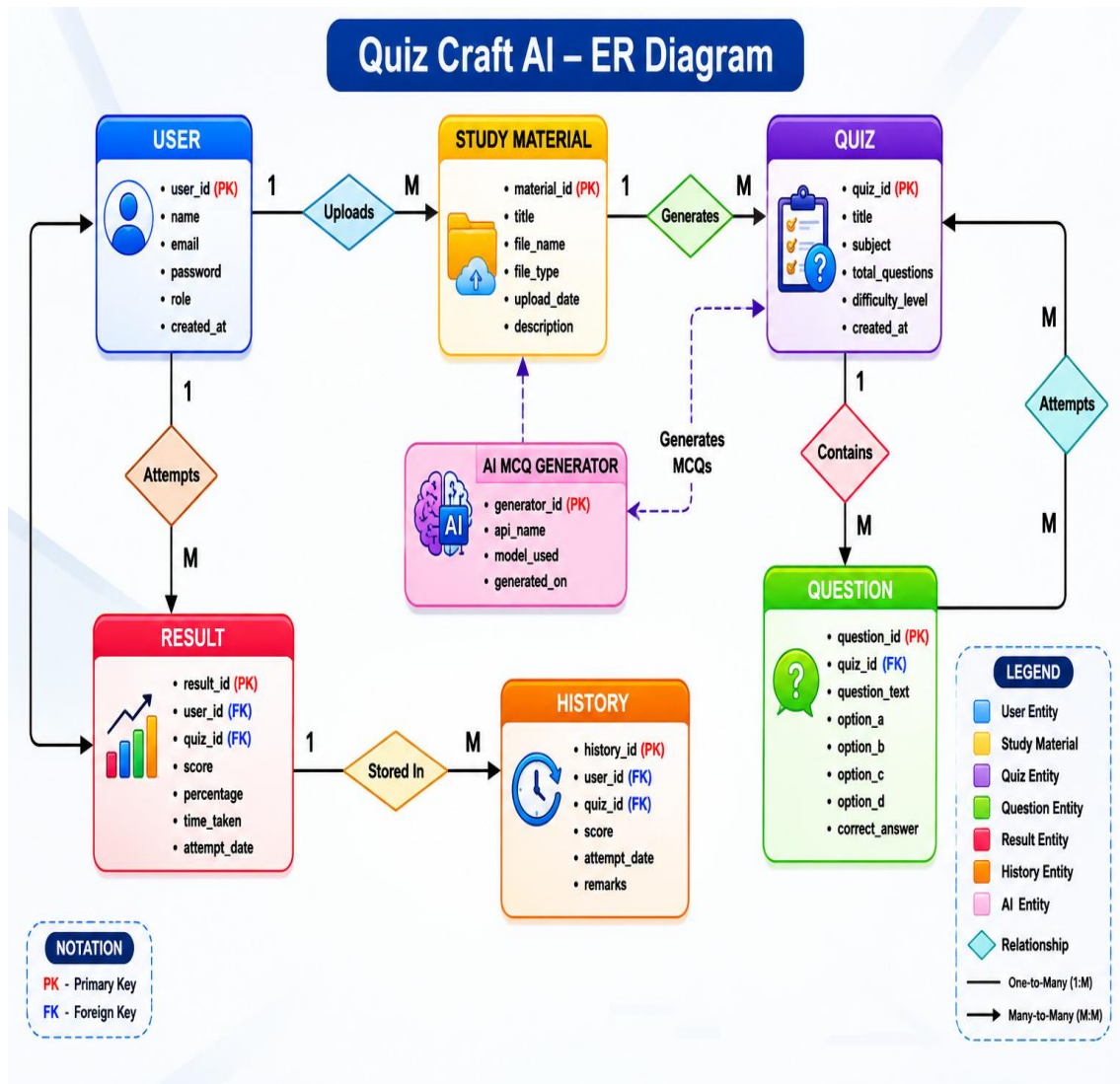


Figure 3.6: ER Diagram of Quiz Craft AI

3.6 Methodology

The working methodology of Quiz Craft AI describes the sequence of operations performed by the system to generate AI-powered multiple-choice quizzes from educational content. The methodology ensures efficient processing of study materials while maintaining accuracy, reliability, and usability.

The overall workflow consists of the following stages:

1. User Authentication

The process begins with user authentication. Users can register by providing their details and creating login credentials. Existing users can access the system through the login interface. Authentication ensures that only authorized users can upload study materials, generate quizzes, access results, and track learning progress.

2. Study Material Upload

After successful authentication, users can upload study materials in PDF or document format. The upload interface validates the selected file and securely transfers it to the backend server. Uploaded materials are stored temporarily for further processing and AI-based analysis.

3. Content Processing

The uploaded study material is processed by the backend system. Important textual information is extracted from the uploaded document and prepared for AI analysis. The content processing stage ensures that the educational material is converted into a format suitable for question generation.

4. AI MCQ Generation

The processed content is forwarded to the AI MCQ Generation Module. The AI system analyzes the educational material, identifies important concepts, topics, and keywords, and automatically generates multiple-choice questions along with answer options and correct answers.

5. Quiz Creation

The generated MCQs are organized into a structured quiz. Quiz information, questions, and metadata are stored in the MongoDB database. The system may also apply question randomization to ensure fairness during assessment.

6. Quiz Attempt

The generated quiz is displayed to the user through the frontend interface. Users can attempt the quiz by selecting answers for each question and submitting their responses upon completion.

7. Result Evaluation

After quiz submission, the system evaluates the user responses by comparing them with the correct answers stored in the database. The evaluation module calculates the final score, percentage, and performance summary automatically.

8. Dashboard Update

The generated results are stored in the MongoDB database and displayed on the user dashboard. The dashboard provides visual access to quiz scores, performance statistics, generated quizzes, and learning progress reports.

9. Historical Tracking

Quiz Craft AI maintains records of previously generated quizzes and assessment results. Historical tracking enables users to review past quiz attempts, compare performance, and monitor learning progress over time.

Methodology Workflow Summary

The complete workflow of Quiz Craft AI can be summarized as follows:

User Authentication → Study Material Upload → Content Processing → AI MCQ Generation → Quiz Creation → Quiz Attempt → Result Evaluation → Dashboard Update → Historical Tracking

This methodology enables Quiz Craft AI to efficiently process educational content, generate meaningful AI-powered quizzes, evaluate user performance, and provide an interactive learning and assessment platform.

Chapter 4

Implementation, Testing, and Maintenance

4.1 Introduction to Languages, IDEs, Tools and Technologies Used for Implementation

Technologies Used

The Quiz Craft AI system was developed using modern web technologies and Artificial Intelligence services to provide an efficient and scalable platform for automatic MCQ generation and online assessment.

Programming Languages Used

- JavaScript: Used for frontend and backend development.
- HTML5: Used for structuring web pages.
- CSS3: Used for styling and responsive design.
- JSON: Used for data exchange between client and server.

Frontend Technologies

- React.js: Used to build a dynamic and interactive user interface.
- Tailwind CSS: Used for responsive and modern UI design.
- JavaScript: Used for handling client-side interactions and quiz functionalities.

Backend Technologies

- Node.js: Provides the runtime environment for server-side execution.
- Express.js: Handles routing, API creation, and middleware management.
- JWT Authentication: Used for secure user authentication and authorization.
- Multer: Used for handling file uploads such as PDF study materials.

Database

- MongoDB: Stores user profiles, uploaded study materials, generated quizzes, questions, results, and historical records.

Artificial Intelligence Technology

- Google Gemini API: Used for analyzing uploaded study materials and generating intelligent multiple-choice questions automatically.
- AI Content Processing Module: Extracts important concepts, topics, and keywords from educational content for quiz generation.

IDEs and Development Tools

- Visual Studio Code (VS Code): Primary development environment.
- Git: Version control system.
- GitHub: Source code repository and collaboration platform.
- Postman: API testing and debugging tool.
- MongoDB Compass: Database management and visualization tool.

Deployment Platforms

- Vercel: Frontend hosting and deployment.
- Render: Backend hosting and deployment.
- Cloud Services: Used for AI processing and application hosting.

Technology Workflow Summary

Study Material Upload → Content Processing → Google Gemini API → MCQ Generation → MongoDB Storage → Quiz Interface → Result Evaluation → Dashboard

4.3 Project Scheduling Using Gantt Chart and PERT Analysis

The project followed the Agile Software Development Life Cycle (SDLC) model. Development activities were divided into multiple phases to ensure systematic planning, implementation, testing, and deployment of the Quiz Craft AI system.

Project Schedule

Phase	Activity	Duration
1	Requirement Analysis	1 Week
2	System Design & Planning	1 Week
3	Frontend Development	2 Weeks
4	Backend Development	2 Weeks
5	Database Integration	1 Week
6	AI MCQ Generation Module Integration	1 Week

Phase Activity	Duration
7 Dashboard & Analytics Development	1 Week
8 Testing & Debugging	1 Week
9 Deployment & Documentation	1 Week

Gantt Chart Representation

Requirement Analysis

Design

Frontend Development

Backend Development

Database Integration

AI Module Integration

Dashboard Development

Testing

Deployment & Documentation

PERT Analysis

The major project activities were planned sequentially:

Requirement Analysis → System Design → Frontend Development → Backend Development → Database Integration → AI MCQ Generation Module Integration → Dashboard Development → Testing → Deployment

This scheduling approach helped in monitoring project progress, tracking milestones, managing development activities efficiently, and ensuring timely completion of the Quiz Craft AI project.

4.4 Testing Techniques and Test Plans

Testing was performed to ensure the reliability, performance, usability, and correctness of the Quiz Craft AI system.

Testing Techniques Used

1. Unit Testing

Individual modules such as user authentication, file upload, AI MCQ generation, quiz management, and database operations were tested independently to verify correct functionality.

2. Integration Testing

Interactions between the frontend, backend, MongoDB database, and AI Question Generation Module were tested to ensure smooth communication and proper data flow.

3. System Testing

The complete Quiz Craft AI application was tested as a single integrated system to verify overall functionality and system behavior.

4. Functional Testing

All functional requirements such as study material upload, AI-based MCQ generation, quiz creation, quiz participation, result evaluation, and dashboard visualization were verified.

5. User Interface Testing

The application's responsiveness, usability, navigation, and user experience were tested on different screen sizes and web browsers.

6. Performance Testing

System response time was evaluated for study material uploads, AI-generated quiz creation, quiz loading, and result generation to ensure efficient performance.

Maintenance Strategy

The system follows corrective, adaptive, and perfective maintenance approaches:

- **Corrective Maintenance:** Fixes bugs, logical errors, and system issues identified after deployment.
- **Adaptive Maintenance:** Updates compatibility with new AI services, APIs, frameworks, and technologies.
- **Perfective Maintenance:** Improves system performance, user interface design, and user experience based on feedback and future requirements.

Regular database backups, dependency updates, security improvements, and AI model optimizations are performed to ensure long-term reliability, scalability, and security of the Quiz Craft AI platform

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 User Interface Representation of Quiz Craft AI

The Quiz Craft AI system provides a modern, responsive, and user-friendly interface for uploading study materials, generating AI-powered multiple-choice quizzes, attempting assessments, and tracking learning progress. The application is designed to simplify quiz creation through Artificial Intelligence while providing an interactive learning experience for students and educators.

The interface consists of the following major modules:

5.1.1 Brief Description of Various Modules of the System

1. Home Module

The Home Module serves as the primary entry point of the application. Users can access information about the platform, explore its features, and navigate to major functionalities such as study material upload, quiz generation, dashboard access, and user authentication. The module provides a simple and intuitive user experience.

2. Study Material Upload Module

This module allows users to upload educational content such as PDF documents, notes, textbooks, and learning materials. The uploaded content serves as the primary input for AI-powered question generation. The interface supports file validation and secure content handling.

3. AI MCQ Generation Module

After the study material is uploaded, the AI MCQ Generation Module analyzes the content and automatically generates multiple-choice questions. The generated questions include answer options and correct answers. This module eliminates the need for manual quiz creation and improves assessment efficiency.

4. Quiz Module

The Quiz Module presents AI-generated questions in a structured and user-friendly format. Users can attempt quizzes, select answers, navigate between questions, and submit assessments. The system may also randomize questions to ensure fairness during evaluations.

5. Dashboard Module

The Dashboard Module provides a centralized view of generated quizzes, assessment records, performance statistics, and learning progress. Users can track their scores, view recent quiz attempts, and monitor their overall performance through graphical and statistical representations.

6. Result Evaluation Module

This module displays quiz outcomes and performance summaries after assessment completion. It calculates scores automatically and presents detailed feedback, including correct and incorrect responses, percentages, and performance insights.

7. History Module

The History Module stores records of previously generated quizzes, attempted assessments, and obtained scores. Users can review earlier quiz attempts, compare results, and monitor their learning progress over time.

8. User Profile Management Module

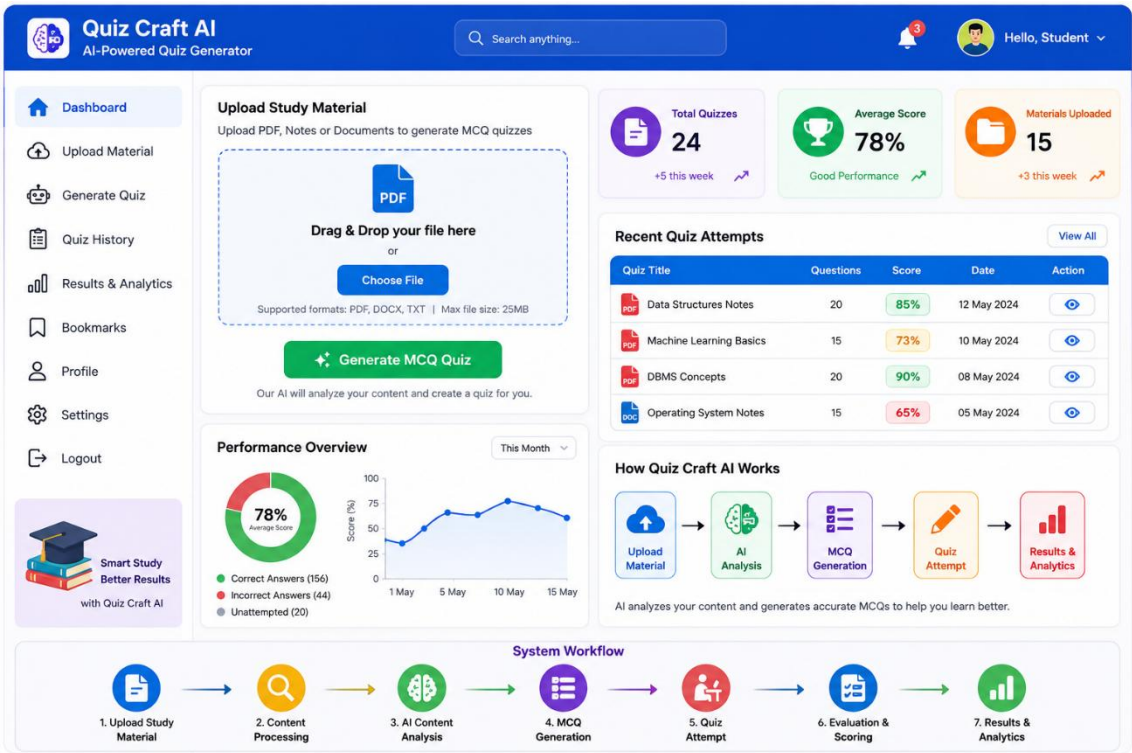
The Profile Module allows users to manage account information, update profile details, access quiz history, and customize preferences. It provides a centralized location for managing user-related information and supports future system enhancements.

9. Authentication Module

This module manages user registration, login, logout, and account security. It ensures that only authorized users can access study materials, generated quizzes, and performance records.

The modular user interface design of Quiz Craft AI ensures ease of use, efficient navigation, and seamless interaction between users and AI-powered quiz generation functionalities, making the platform suitable for modern educational and assessment environments.

5.2 Snapshots of System with Brief Detail of Each



Description:

The Home Dashboard serves as the main entry point of the Quiz Craft AI system. It provides users with an intuitive and user-friendly interface for uploading study materials such as PDF documents, notes, and educational content. Users can upload files using drag-and-drop functionality or by selecting files manually from their device.

After the upload process is completed, the system validates the file and prepares it for AI-based content analysis. The dashboard also provides quick access to major system functionalities such as quiz generation, performance tracking, quiz history, and profile management.

This interface acts as the starting point of the Quiz Craft AI workflow and enables users to efficiently transform study materials into automatically generated multiple-choice quizzes. The responsive design ensures smooth operation across desktops, tablets, and mobile devices.

Figure 5.2: AI Generated MCQs and Quiz Interface

The screenshot displays the Quiz Craft AI interface, which is designed for generating and taking multiple-choice quizzes. The top navigation bar includes the Quiz Craft AI logo, a title 'AI Generated MCQs', and user statistics (Score: 12/20, Progress: 60%). A sidebar on the left provides navigation options like Dashboard, Upload Material, Generate Quiz, Quiz History, Results & Analytics, Bookmarks, Profile, Settings, and Logout. The main content area shows a quiz titled 'Data Structures and Algorithms' with 20 questions of Medium difficulty. Two sample questions are visible: 'What is the time complexity of binary search in a sorted array?' and 'Which data structure follows the FIFO (First In First Out) principle?'. Both have 'B. O(log n)' and 'B. Queue' as the selected correct answers. A 'Quiz Information' panel on the right shows the Quiz ID, Start Time, Time Elapsed, and Total Marks. A 'Question Navigation' grid allows users to jump between questions. A 'Quick Tip' and 'AI Generated' notice are also present. At the bottom, a 'Quiz Craft AI Workflow' diagram illustrates the process: 1. Upload Study Material, 2. AI Analysis of Content, 3. MCQ Generation, 4. Quiz Attempt, and 5. Result & Performance.

Description:

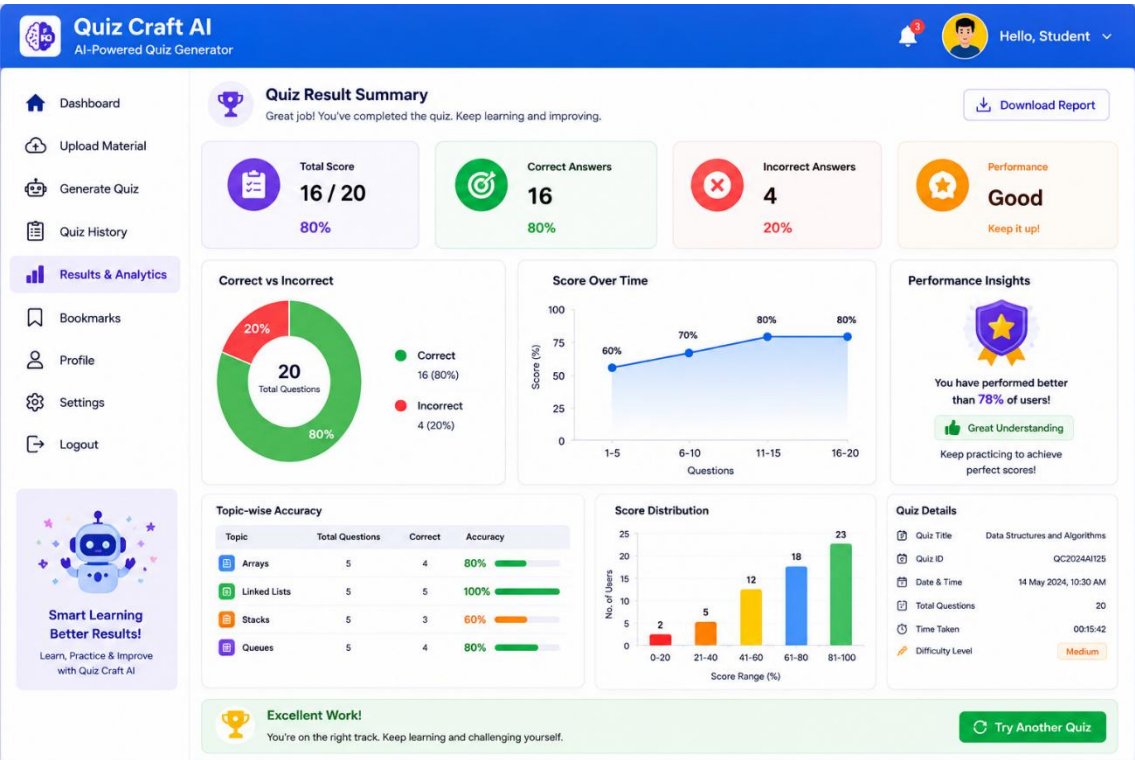
This screen displays the multiple-choice questions automatically generated from the uploaded study material using Artificial Intelligence. The interface presents each question along with multiple answer options in a structured and user-friendly format.

Users can attempt the quiz by selecting appropriate answers and navigating through the generated questions. The system organizes questions based on the analyzed content and ensures that they are relevant to the uploaded educational material.

The interface also displays quiz-related information such as total questions, difficulty level, progress status, and remaining questions. This module serves as the core assessment component of Quiz Craft AI and enables users to evaluate their understanding of the study material efficiently.

The AI-generated questions help reduce manual effort in quiz creation while providing an interactive and effective learning experience for students.

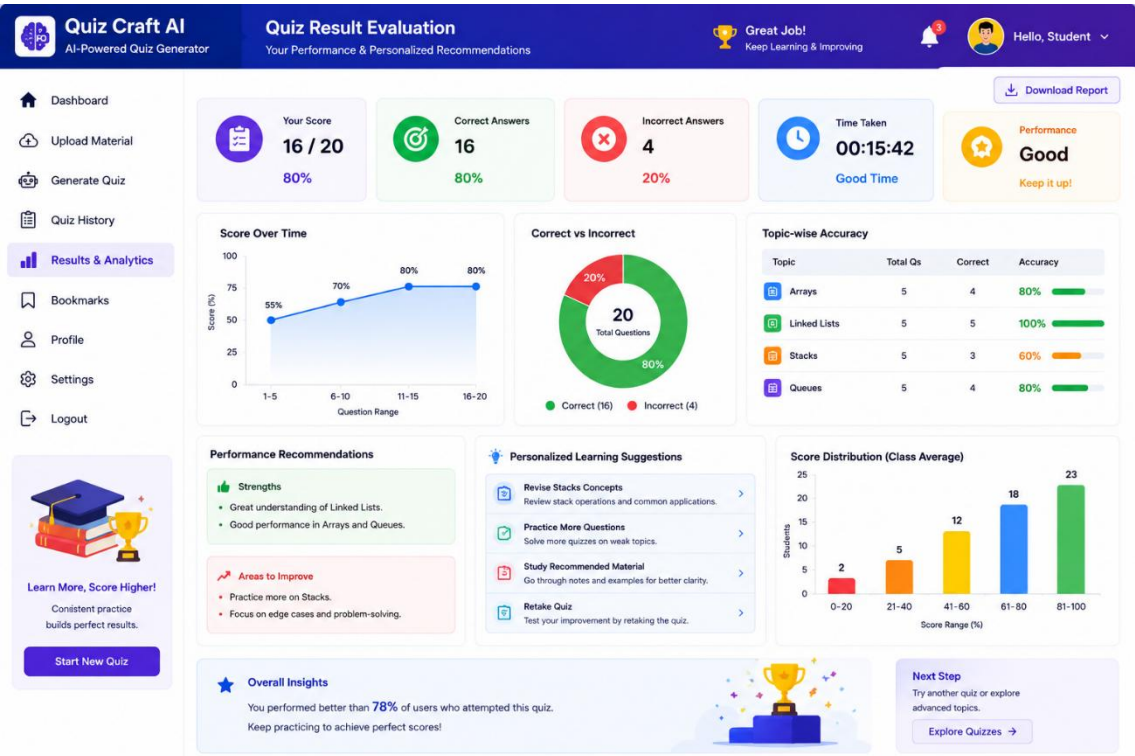
Figure 5.3: AI Synthesis Summary and Risk Profile



Description:

This screen displays the overall performance summary after quiz completion. The system automatically evaluates user responses and generates detailed analytics including total score, percentage, correct answers, incorrect answers, and overall performance metrics. The Performance Analytics section provides visual insights through charts and statistics, helping users identify their strengths and weaknesses. Users can review their quiz performance, monitor learning progress, and analyze topic-wise accuracy levels. The AI-powered evaluation system generates instant feedback and performance reports, enabling students to improve their understanding of the uploaded study material. This module plays an important role in self-assessment and continuous learning within the Quiz Craft AI platform.

Figure 5.4: Personalized Dietary and Clinical



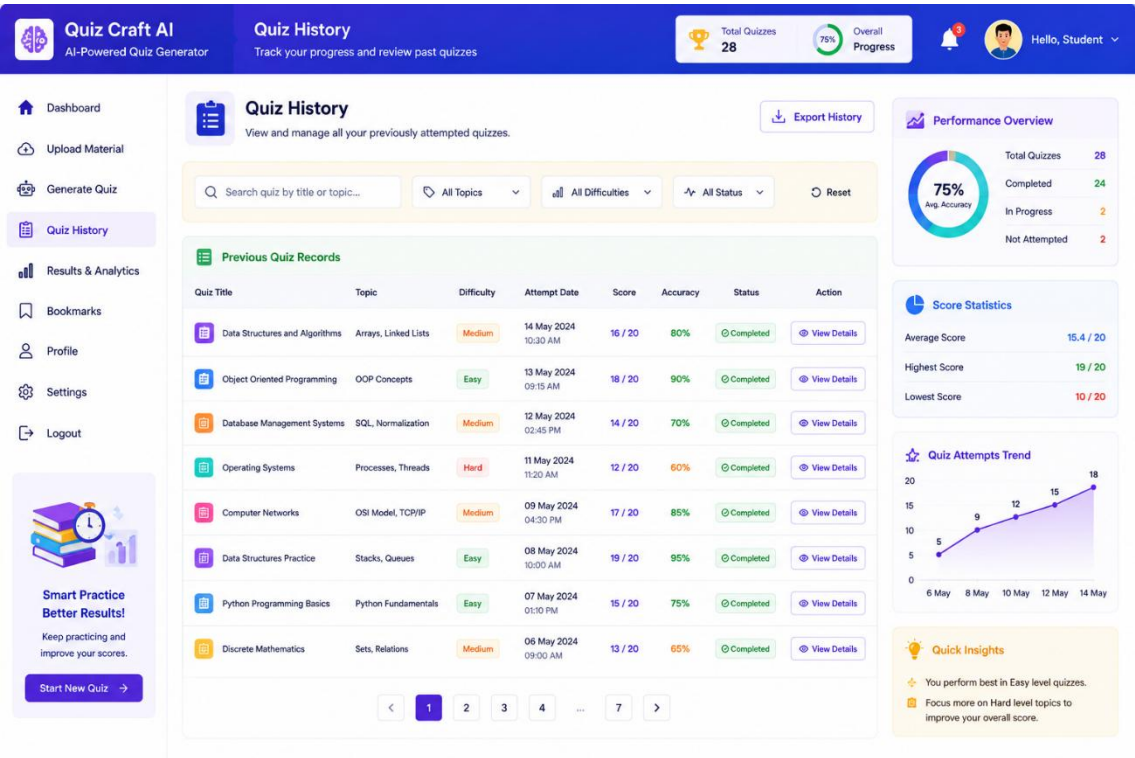
Description:

This screen displays the detailed quiz evaluation results generated after assessment completion. The system automatically analyzes user responses and provides performance insights based on the obtained score, accuracy percentage, and topic-wise understanding. The Performance Recommendation section highlights weak and strong areas identified during the quiz attempt. Based on the evaluation results, the system provides personalized learning recommendations, suggested topics for revision, and improvement strategies to enhance academic performance.

The interface also presents topic-wise accuracy, score distribution, performance trends, and learning analytics through graphical visualizations. These insights help students identify knowledge gaps, focus on important concepts, and improve their overall learning effectiveness.

This module enhances self-assessment by transforming quiz results into actionable learning recommendations, making Quiz Craft AI an intelligent educational assessment platform.

Figure 5.5: History Management Module



Description:

The History Management Module stores all previously generated quizzes, quiz attempts, scores, and performance records. It enables users to access past assessments, review quiz results, and monitor their learning progress over time.

The module provides an organized view of historical quiz data, including quiz titles, attempt dates, scores obtained, accuracy percentages, and completion status. Users can search, filter, and revisit previously attempted quizzes whenever required.

The History Module helps students track their academic performance, identify improvement areas, and compare results across multiple quiz attempts. By maintaining a complete record of learning activities, the system supports continuous self-assessment and effective progress monitoring.

This feature enhances the overall learning experience by providing easy access to historical assessment data and personalized performance tracking within the Quiz Craft AI platform.

CHAPTER 6

CONCLUSION

6.1 Conclusion

The Quiz Craft AI – AI Driven MCQ Quiz Generator successfully addresses one of the major challenges in modern education: creating assessment questions efficiently from study materials. Traditional quiz creation requires significant time and effort from educators, while students often lack sufficient practice resources. Quiz Craft AI bridges this gap by automatically generating multiple-choice questions from educational content using Artificial Intelligence.

The system combines modern web technologies with Artificial Intelligence to provide an efficient platform for quiz generation and online assessment. Through the integration of React.js, Node.js, Express.js, MongoDB, and Google Gemini API, the application is capable of processing uploaded study materials and generating meaningful MCQs automatically. The system analyzes educational content, extracts important concepts, and transforms them into structured assessment questions.

During implementation and testing, the system demonstrated reliable performance in study material processing, AI-based question generation, quiz management, and result evaluation. The generated quizzes helped reduce manual effort while providing students with a convenient method for self-assessment and practice. The dashboard, performance tracking features, and quiz history management further enhanced the overall user experience.

The project also highlights the potential of Artificial Intelligence in educational technology. By automating quiz creation and providing instant assessment capabilities, Quiz Craft AI promotes effective learning and improves accessibility to educational resources. The successful implementation of the system confirms the feasibility and effectiveness of AI-powered assessment solutions.

In conclusion, Quiz Craft AI is a practical, scalable, and user-friendly educational platform. It simplifies quiz generation, supports online learning, improves assessment efficiency, and demonstrates how AI can be effectively utilized in the education sector.

The project provides a strong foundation for future enhancements and large-scale deployment in academic institutions and online learning environments.

6.2 Future Scope

Future versions of Quiz Craft AI can be enhanced with adaptive learning capabilities, allowing the system to generate quizzes based on a student's previous performance and learning patterns. This would create a more personalized learning experience and improve assessment effectiveness.

The platform can also be extended with dedicated mobile applications for Android and iOS devices, enabling users to generate quizzes and practice assessments anytime and anywhere.

Additional features such as difficulty-level selection, topic-wise quiz generation, automated question categorization, and advanced performance analytics can further improve usability. Integration with Learning Management Systems (LMS) can also make the platform suitable for schools, colleges, coaching institutes, and corporate training programs.

Advanced AI models may be incorporated to generate higher-quality questions, provide detailed explanations for answers, and recommend personalized learning resources based on user performance. Features such as voice-based interaction, multilingual quiz generation, and collaborative learning environments can further enhance the platform.

Integration with cloud-based educational platforms and online examination systems can enable large-scale deployment and centralized management of assessments. Real-time analytics and predictive performance tracking can also help educators identify learning gaps and improve teaching strategies.

These enhancements would transform Quiz Craft AI from an AI-powered MCQ generator into a comprehensive intelligent learning and assessment platform capable of supporting students, educators, and institutions throughout the learning process.